[opcode | add1]

1. 1 address instructions are supported in

accumulator

a. AC -based architecture

b. Register based architecture

c. stack-based architecture

d. Register-memory based architecture

Ans: a, b, d

AC based architecture only supports 1 address instruction

Register based architecture can have 1 address and 2 address instruction.

stack-based architecture only supports 0 address instruction.

Register-mly → can support 0, 1, 2, 3 address instrn

2. Which of the following cache mapping(s) may experience Conflict miss?

a) Direct mapping

b) 2-way set associative mapping

c) 4-way set associative mapping

d) Fully associative mapping

Ans: a, b, c

Cache miss - occur when data is not available in cache mly.

Types of cache Misses

1. Compulsory Miss (cold start miss or first reference miss)

These misses occur when the first access to a block happens. Block must be brought into the cache. [occur in 3 cache mapping techniques]

2. Capacity Miss low in direct, medium in set associative, high in associative

These misses occur when the program working set is much larger than the cache capacity. Cache cannot contain all blocks needed for program execution, so cache discards these blocks.

3. Conflict Miss (collision miss or Interference miss)

These misses occur when several blocks are mapped to the same set or block frame. These misses occur in the direct high in mapped or medium in set associative mapped block placement strategies.

4. Coherence Miss (Invalidation)

These misses occur when other external processors ie I/O updates memory. [occur in 3 cache mapping techniques.]

3. The basic principle underlying behind the concept of cache memory is:-

a. stored program concept
b. Locality of reference
c. Divide and conquer
d. NOTA

Ans: b

Locality of reference

temporal	spatial
* Recently executed instruction is likely to be executed again very soon	* Instructions with addresses close to a recently used instruction are likely to be executed soon.

4. Consider a direct mapped cache of size 256 Kilo words with block size 512 words. There are 6 bits in the tag. The no: of bits in block (index) and word (offset) fields of physical address are is:

a) block (index) field = 6 bits, word (offset) field = 9 bits
b) block (index) field = 7 bits, word (offset) field = 8 bits
c) block (index) field = 9 bits, word (offset) field = 9 bits
d) block (index) field = 8 bits, word (offset) field = 8 bits

Ans: C

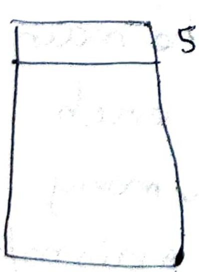
$$\frac{\text{cache size}}{\text{Line size}} = \frac{256\ KB}{512\ \text{Bytes}}$$

$$= \frac{256 \times 1024}{512}$$

$$= \frac{2^8 \times 2^{10}}{2^9}$$

$$= 2^9 \text{ bytes}$$

$$= \underline{\underline{9 \text{ bits}}}$$



512 words or bytes

256 Kilo words or Kilo Bytes

tag	Block	Word
6 bits		

→ Block size = ~~256 KB~~ 512 bytes
= ~~256~~ = 2^9 bytes
∴ 9 bits

5 The cache memory of 1K words uses direct mapping with a block size of 4 words. How many blocks can the cache accomodate?

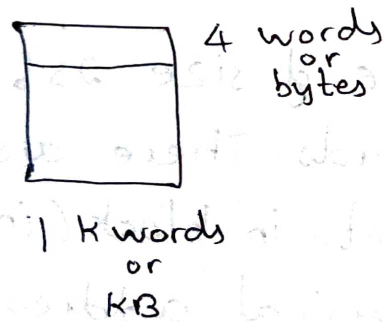
a. 256 words

b. 512 words

c. 1024 words

Ans: a

d. 128 words



$$\text{No: of blocks} = \frac{1\ KB}{4\ bytes}$$

$$= \frac{1 \times 1024\ bytes}{4\ bytes}$$

$$= \frac{2^{10}\ bytes}{2^{2}\ bytes}$$

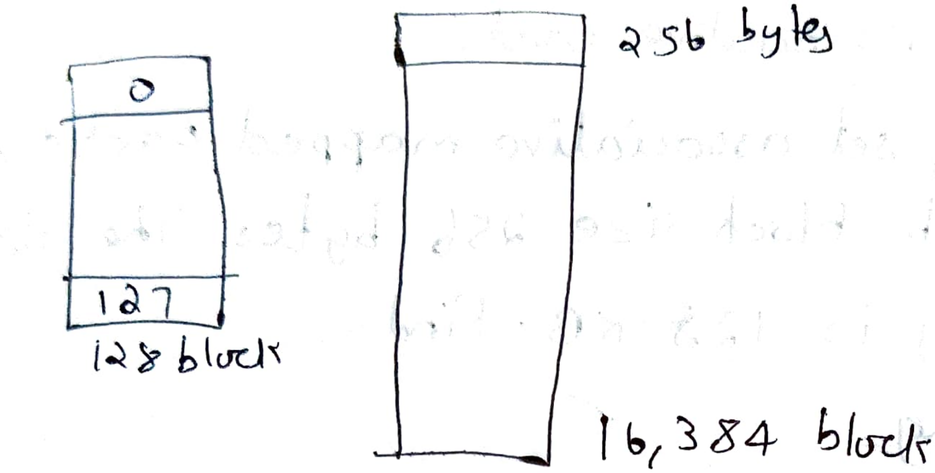
$$= 2^{8}\ \text{lines}$$

$$= 256\ \text{words}$$

6. A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 8-bit words. How many bits are required for addressing the main m/y?

a) 22 b) 20 c) 32 d) 36

Ans: a



No: of bits in physical address?

1 block = 256 bytes

16,384 block = ? bytes

$= 16,384 \times 256$

$= 2^{14} \times 2^{8}$

$= 2^{22}$ bytes

$\therefore$ No: of bits in physical address = 22 bits

7. SCSI stands for

a. Simple computer serial interface
b. Small computer system interface
c. Serial controller for system interface.
d. Simple computer serial interface.

Ans: b It is a set of standards for physically connecting and transferring data between computers and peripheral devices (CD-ROM drives, scanners, DVD drives and CD writers.

8. A static RAM cell contains

a. Transistor
b. Capacitor
c. Inverter
d. Register

Ans: a

9. PLA means

a) Programmed Large Array
b) Programmable Logic Array
c) Programmed Long Array
d) Programmable List Array

Ans: b.

10. A microprogram sequencer

a. generates the address of next microinstruction to be executed.
b. generates the control signals to execute a micro instruction
c. Sequentially averages all microinstructions in the control m/y.

d. enables the efficient handling of a micro program subroutine.

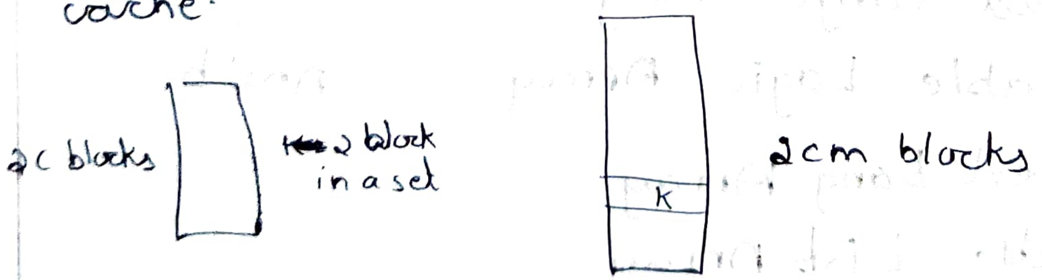
Ans: a

11. The main memory of a computer has 2cm blocks while the cache has 2c blocks. If the cache uses the set associative mapping scheme with 2 blocks per set, then block k of the main mly maps to the set:

1. (k mod m) of the cache
2. (k mod c) of the cache
3. (k mod 2c) of the cache
4. (k mod 2cm) of the cache

Ans: k mod c of the cache

Block j of main memory maps to set number (j modulo number of sets in the cache) of the cache.



Block k of main mly maps to k mod no: of sets in the cache

No: of sets in the cache = No: of blocks in cache / No: of blocks in one set

$= \frac{2c}{2}$

$= \underline{\underline{c}}$

Ans: is k mod c of the cache.

12. A 32-bit processor has an addressable space of ___ Giga bytes.

a. 4
b. 2
c. 1
d. NOTA

Ans: 4GB

$$\frac{2^{32}}{1024 \times 1024 \times 1024} = \frac{2^{32}}{2^{10} \times 2^{10} \times 2^{10}} = \frac{2^{32}}{2^{30}} = 2^2 = 4 \text{ GB}$$

$2^{32} = 4,294,967,296$ bytes

$\Rightarrow \frac{4,294,967,296}{1024} = 4,194,304$ KB

$\Rightarrow \frac{4,194,304}{1024} = 4096$ MB

$\Rightarrow \frac{4096}{1024} = 4$ GB

13. In Reverse Polish Notation (ie Postfix notation), expression A*B+C*D is written as.

a. AB*CD*+
b. A*BCD*+
c. AB*CD+*C
d. A*B*CD+

Ans: a

Ans: Given is an infix expression so convert it to postfix.
Reverse Polish Notation is a method for conveying mathematical expressions without the use of separators such as brackets.

$A * B + C * D$

$AB*CD*+$

14. SIMD represents an organization that

a. refers to a computer system capable of processing several programs at the same time.

b. represents organization of single computer containing a control unit, processor unit and a memory unit.

c. includes many processing units under the supervision of a common control unit.

d. NOTA.

Ans: c   single instruction/Multiple data

15. Floating point representation is used to store

a. Boolean values

b. whole numbers

c. real integers — Ans: c

d. integers

16. Assembly language

a. uses alphabetic codes in place of binary numbers used in machine language.

b. is the easiest language to write programs

c. need not be translated into machine language

d. NOTA

Ans: a

17. In computers, subtraction is generally carried out by.

a. 9's complement

b. 10's "

c. 1's "

d. 2's "

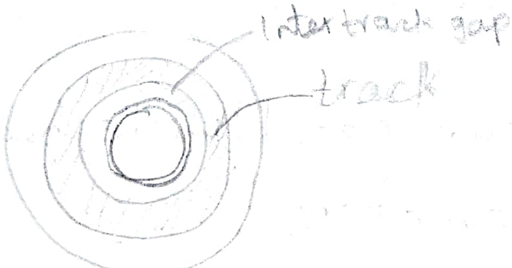
Ans: 2's complement.

Magnetic Disk

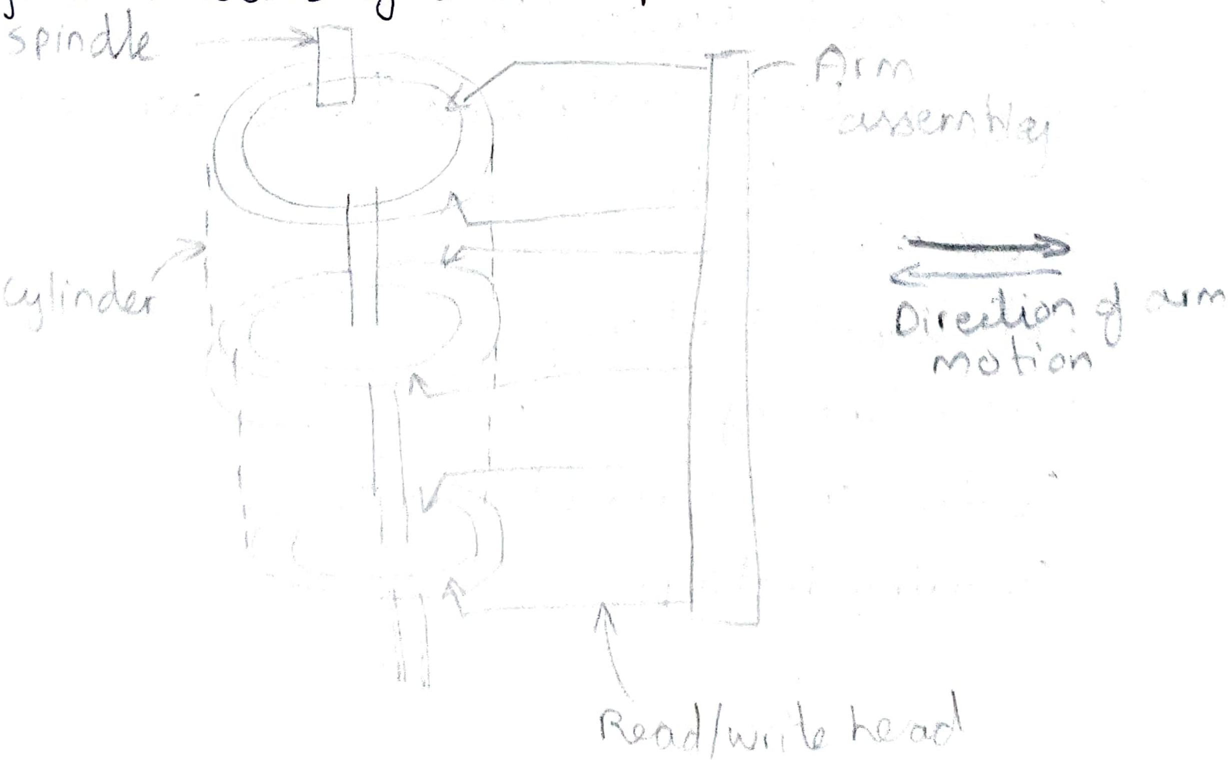
* Magnetic Disk is a storage device that is used to write, rewrite and access data.
* It uses a magnetization process

Architecture

* The entire disk is divided into platters
* Each platter consists of concentric ~~co~~ circles called as tracks
* These tracks are further divided into sectors which are the smallest divisions in the disk.



* A cylinder is formed by combining the tracks at a given radius of a disk pack.



* There exists a mechanical arm called as Read/write head.
* It is used to read from and write to the disk.
* Head has to reach ~~&~~ at a particular track and then wait for the rotation of the platter
* The rotation causes the required sector of the track to come under the head.
* Each platter has 2 surface - top and bottom and both surfaces are used to store data.
* Each surface has its own read/write head.

Disk Performance Parameters

* The time taken by the disk to complete an I/O request is called as disk access time or disk service time
* Components that contribute to the service time are:-

Seek time
Rotational Latency
Data Transfer Rate
Controller Overhead
Queuing Delay.

1. Seek time

The time taken by the read/write head to reach the desired track is called as seek time.

* It is the component which contributes the largest percentage of the disk service time.
* The lower the seek time, the faster the I/O operation.
* Seek time specifications include

a) Full stroke :- It is the time taken by the read/write head to move across the entire width of the disk from the innermost track to the outermost track.

b) Average seek time:- It is the average time taken by the read/write head to move from one random track to another.

Average seek time = $\frac{1}{3}$ × Full stroke

c) Track to Track :- It is the time taken by the read/write head to move b/w the adjacent tracks.

2. Rotational Latency :- ~~The~~

The time taken by the desired sector to come under the read/write head is called as rotational Latency.

* It depends on rotation speed of the spindle.

Average rotational latency $= \frac{1}{2} \times$ Time taken for full rotation.

3. Data Transfer Rate

* The amount of data that passes under the read/write head in a given amount of time is called as data transfer rate.
* The time taken to transfer the data is called as transfer time

It depends on the following factors

1) No: of bytes to be transferred
2) Rotation speed of the disk
3) Density of the track
4) Speed of the electronics that connects the disk to the computer.

4. Controller Overhead

The overhead imposed by the disk controller is called as controller overhead.

Disk controller is a device that manages the disk.

5) Queuing Delay

The time spent waiting for the disk to become free is called as queuing delay.